

Project #2: Application of Curve Fitting Algorithms

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Problem Specification:

The whole purpose of this project is to use curve fitting methods to model the relationship between the thermal conductivity of copper and temperature. Thermal conductivity describes a material's capability to transfer heat. Since the thermal conductivity of a material can change with temperature, experimental data can be used to develop mathematical models that predict the relationship.

For this project, experimental temperature and thermal conductivity data for copper were looked at using two different model equations. Each equation was fit to the experimental data using General Linear Least Squares (GLLS) and MATLABs fminsearch method. This process resulted in four different curve fits. The quality of each fit was evaluated using coefficient of determination (r^2) and standard error. The Results were compared to decide which method and equation provided the best fit to the experimental data.

Mathematic formulation:

There were two model equations that were used to fit the experimental thermal conductivity data. The first equation was:

$$k = \frac{c_1}{c_2 T^2}$$

Where K represented the thermal conductivity, T represents temperature, and c1 and c2 are the coefficients being determined. To use GLLS, I took the reciprocal of both sides to linearize it.

$$\frac{1}{k} = \frac{c_1}{T} + c_2 T^2$$

This made it so the coefficients c1 and c2 to be determined using linear least-square solution. The second model equation was:

$$k = \frac{1}{\frac{c_1}{T} + c_2 T^2 + c_3 T^2}$$

Equation 2 contains three coefficients, c1, c2, c3. IT had the same thing done to it as the first equation:

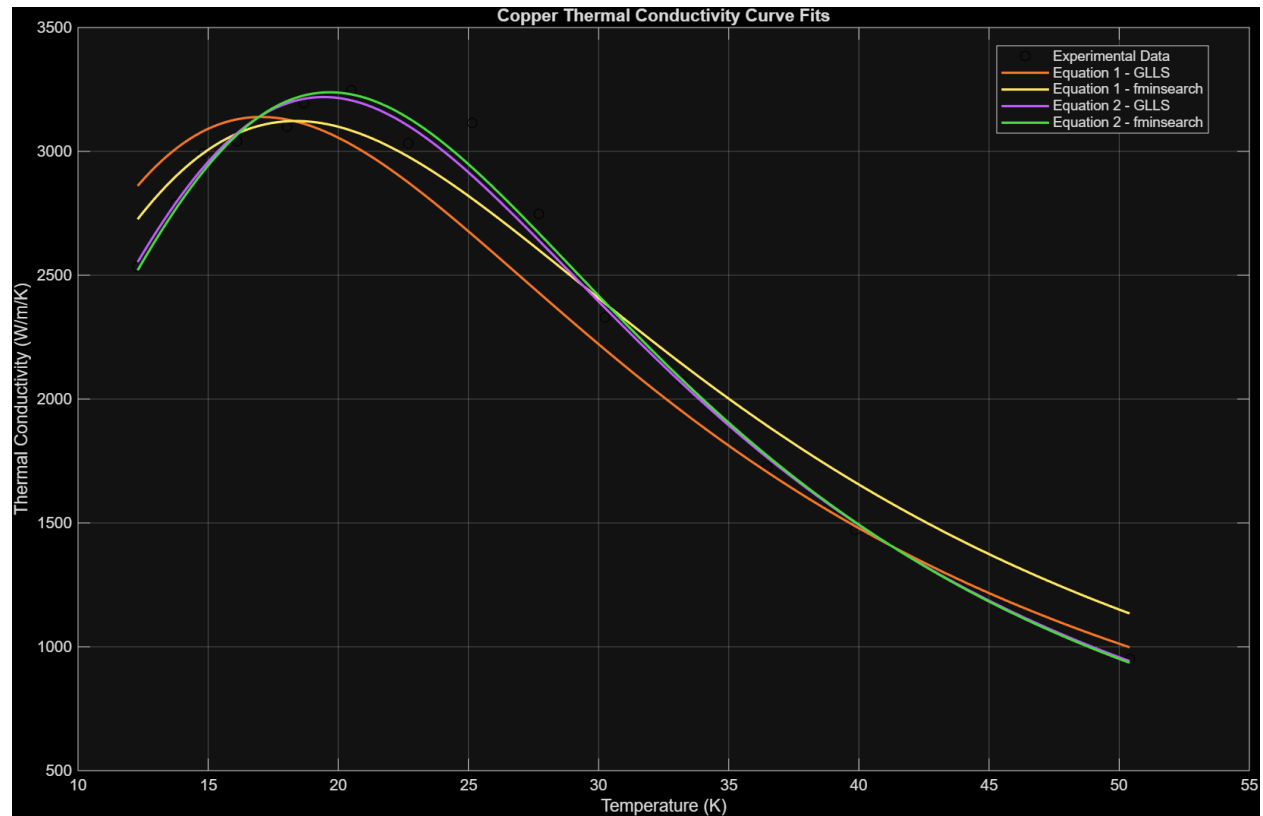
$$\frac{1}{k} = \frac{c_1}{T} + c_2 T^2 + c_3 T^2$$

To calculate the coefficients for both equations, GLLS was used. MATLABs' fminsearch method was also used for both equations. For fminsearch, the coefficients were adjusted to minimize the sum of the squared residuals between the measured and predicted thermal conductivity values.

The quality of each curve fit was evaluated using the sum of squared residuals, standard error, coefficient of determination (r^2), and total sum of squares. The smaller a standard error is, the less error between the model and the experimental data, while an r^2 value closer to 1 indicates the best fit.

Figure:

Figure 1. Copper Thermal Conductivity Curve Fits



This figure compares the experimental copper thermal conductivity data with all four of the calculated curve fits. The equation 2 curves follow the experimental data more closely than the equation 1 curves over the temperature range. The GLLS and fminsearch curves for equation 2 are decently close to one another.

Tables:

Table 1. Standard Error and (r^2) Results

Names	SE	r2
"Equation 1 - GLLS"	213.13	0.91532
"Equation 1 - Fminsearch"	153.71	0.95596
"Equation 2 - GLLS"	82.42	0.98831
"Equation 2 - Fminsearch"	79.045	0.98925

Table 2. Curve-Fit Coefficients

Names	c1	c2	c3
"Equation 1 - GLLS"	0.0036168	3.6619e-07	0
"Equation 1 - Fminsearch"	0.0039218	3.1637e-07	0
"Equation 2 - GLLS"	0.0049414	-7.2797e-06	5.238e-07
"Equation 2 - Fminsearch"	0.0051056	-8.1876e-06	5.4346e-07

Results and Discussion:

The results showed that equation 2 fit the experimental data better than equation 1. For equation 1, GLLS had a standard error of 213.13 and an (r^2) value of 0.91532, while fminsearch had a lower standard error of 153.71 and a higher (r^2) value of 0.95596. Equation 2 gave better results with both methods in consideration. GLLS had a standard error of 79.045 and an (r^2) of 0.98925. Since a lower standard error and an (r^2) value closer to 1 indicate a better fit, equation 2 using fminsearch gave the best overall fit to the experimental data.

Conclusion:

For equation 1, fminsearch provide a better fit than GLLS because it had a lower standard error and higher (r^2) value. Fminsearch also performed slightly better for equation 2 for the same reasons. Comparing the two equations, equation 2 was the better model because it produced much lower standard error and (r^2) values closer to 1. Overall, equation 2 using fminseach gave the best fit, with the lowest standard error of 79.045 and the highest (r^2) value of 0.98925.